

Lab Assignment 8

Object-Oriented Programming

Branch
B.E., 2nd Year – ENC



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Function Template

1. Write a function template to swap two variables of any data type.

```
#include <iostream>
using namespace std;

template <class T>
void swapValues(T &a, T &b) {
    T temp;
    temp = a;
    a = b;
    b = temp;
}

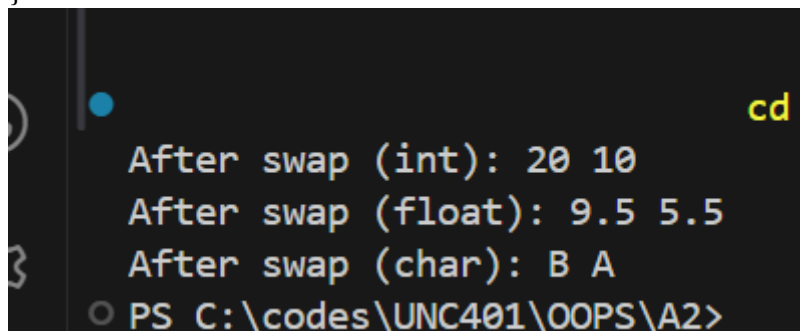
int main() {

    int x = 10, y = 20;
    swapValues(x, y);
    cout << "After swap (int): " << x << " " << y << endl;

    float p = 5.5, q = 9.5;
    swapValues(p, q);
    cout << "After swap (float): " << p << " " << q << endl;

    char c1 = 'A', c2 = 'B';
    swapValues(c1, c2);
    cout << "After swap (char): " << c1 << " " << c2 << endl;

    return 0;
}
```



```
cd
After swap (int): 20 10
After swap (float): 9.5 5.5
After swap (char): B A
PS C:\codes\UNC401\OOPS\A2>
```

2. Write a function template to find the minimum element in an array.

```
#include <iostream>
using namespace std;

template <class T>
T findMin(T arr[], int n) {
    T min = arr[0];

    for(int i = 1; i < n; i++) {
        if(arr[i] < min) {
            min = arr[i];
        }
    }

    return min;
}

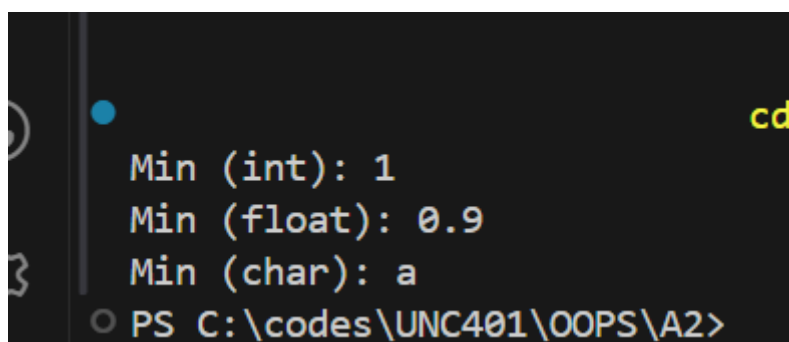
int main() {

    int arr1[] = {5, 2, 8, 1, 9};
    int n1 = 5;
    cout << "Min (int): " << findMin(arr1, n1) << endl;

    float arr2[] = {3.5, 1.2, 4.8, 0.9};
    int n2 = 4;
    cout << "Min (float): " << findMin(arr2, n2) << endl;

    char arr3[] = {'z', 'a', 'm', 'b'};
    int n3 = 4;
    cout << "Min (char): " << findMin(arr3, n3) << endl;

    return 0;
}
```

A screenshot of a terminal window with a dark background. The output of the program is displayed in a light blue monospaced font. The output consists of three lines: "Min (int): 1", "Min (float): 0.9", and "Min (char): a". Below these lines is the command prompt "PS C:\codes\UNC401\OOPS\A2>". In the top right corner of the terminal window, the text "cd" is visible in a yellow font.

```
Min (int): 1
Min (float): 0.9
Min (char): a
PS C:\codes\UNC401\OOPS\A2>
```

3. Write a function template to sort an array using bubble sort.

```
#include <iostream>
using namespace std;

template <class T>
void bubbleSort(T arr[], int n) {
    for(int i = 0; i < n - 1; i++) {
        for(int j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                T temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

template <class T>
void display(T arr[], int n) {
    for(int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

int main() {

    int arr1[] = {5, 3, 8, 1, 2};
    int n1 = 5;

    bubbleSort(arr1, n1);
    cout << "Sorted int array: ";
    display(arr1, n1);

    float arr2[] = {4.5, 2.1, 3.3, 1.0};
    int n2 = 4;

    bubbleSort(arr2, n2);
    cout << "Sorted float array: ";
    display(arr2, n2);

    char arr3[] = {'d', 'a', 'c', 'b'};
    int n3 = 4;

    bubbleSort(arr3, n3);
    cout << "Sorted char array: ";
    display(arr3, n3);

    return 0;
```

```
cd "c:\d
Sorted int array: 1 2 3 5 8
Sorted float array: 1 2.1 3.3 4.5
Sorted char array: a b c d
PS C:\codes\UNC401\OOPS\A2>
```

0 0 Amazon Q Indexing completed.

4. Write a function template to perform linear search.

```
#include <iostream>
using namespace std;

template <class T>
int linearSearch(T arr[], int n, T key) {
    for(int i = 0; i < n; i++) {
        if(arr[i] == key) {
            return i;
        }
    }
    return -1;
}

int main() {

    int arr1[] = {5, 3, 8, 1, 2};
    int n1 = 5;
    int key1 = 8;

    int res1 = linearSearch(arr1, n1, key1);
    if(res1 != -1)
        cout << "Element found at index: " << res1 << endl;
    else
        cout << "Element not found" << endl;

    float arr2[] = {2.5, 4.1, 1.3, 3.7};
    int n2 = 4;
    float key2 = 1.3;

    int res2 = linearSearch(arr2, n2, key2);
    if(res2 != -1)
        cout << "Element found at index: " << res2 << endl;
    else
        cout << "Element not found" << endl;

    char arr3[] = {'a', 'b', 'c', 'd'};
    int n3 = 4;
    char key3 = 'c';

    int res3 = linearSearch(arr3, n3, key3);
    if(res3 != -1)
        cout << "Element found at index: " << res3 << endl;
    else
        cout << "Element not found" << endl;

    return 0;
}
```

```
● PS C:\codes> cd "c:\codes\UNC40
Element found at index: 2
Element found at index: 2
Element found at index: 2
○ PS C:\codes\UNC401\OOPS\A2>
```

5. **Design overloaded versions of a function template process() such that:**

i) Accepts a single parameter

ii) Accepts two parameters of the same type

iii) Accepts two parameters of different types

```
#include <iostream>
```

```
using namespace std;
```

```
// Single parameter
```

```
template <typename T>
```

```
void process(T a) {
```

```
    cout << "Single param: " << a << endl;
```

```

    }

    // Two same type
    template <typename T>
    void process(T a, T b) {
        cout << "Same type params: " << a << " " << b << endl;
    }

    // Two different types
    template <typename T, typename U>
    void process(T a, U b) {
        cout << "Different types: " << a << " " << b << endl;
    }

    int main() {
        process(10);
        process(10, 20);
        process(10, 2.5);
        return 0;
    }

```


Output

Clear

```
Single param: 10
Same type params: 10 20
Different types: 10 2.5

=== Code Execution Successful ===
```

Class Template

- 1. Develop a class template for stack with push and pop operations.**

```
#include <iostream>

using namespace std;

template <typename T>

class Stack {

    T arr[100];

    int top;

public:
```

```

Stack() { top = -1; }

void push(T val) {
    arr[++top] = val;
}

void pop() {
    if(top >= 0)
        top--;
}

void display() {
    for(int i = 0; i <= top; i++)
        cout << arr[i] << " ";
}

};

int main() {
    Stack<int> s;

    s.push(10);
    s.push(20);
    s.pop();
    s.display();
}

```

Output

Clear

10

=== Code Execution Successful ===

2. **Develop a class template for queue with enqueue and dequeue operations.**

```
#include <iostream>

using namespace std;

template <typename T>

class Queue {

    T arr[100];

    int front, rear;

public:

    Queue() { front = rear = -1; }
```

```

void enqueue(T val) {
    if(front == -1) front = 0;
    arr[++rear] = val;
}

void dequeue() {
    if(front <= rear)
        front++;
}

void display() {
    for(int i = front; i <= rear; i++)
        cout << arr[i] << " ";
}

};

int main() {
    Queue<int> q;
    q.enqueue(10);
    q.enqueue(20);
    q.dequeue();
    q.display();
}

```

Output

Clear

20

=== Code Execution Successful ===

```
include <iostream> using
namespace std;
template <typename T1, typename T2>
class Pair {
    T1 a;
    T2 b;
```

```
public:
    Pair(T1 x, T2 y) {
        a = x;
        b = y;
    }
    void display() {
        cout << a << " " << b;
    }
};

int main() {
    Pair<int, float> p(10, 2.5);
    p.display();
}
```

Output

Clear

10 2.5

=== Code Execution Successful ===

3. **Create a class template for basic arithmetic operations.**

```
#include <iostream>

using namespace std;

template <typename T>

class Arithmetic {

    T a, b;

public:

    Arithmetic(T x, T y) {

        a = x;

        b = y;

    }

    void add() { cout << "Add: " << a + b << endl; }

    void sub() { cout << "Sub: " << a - b << endl; }

    void mul() { cout << "Mul: " << a * b << endl; }

    void div() { cout << "Div: " << a / b << endl; }

};

int main() {

    Arithmetic<int> obj(10, 5);

    obj.add();

    obj.sub();

    obj.mul();

    obj.div();

}
```

Output

Clear

Add: 15
Sub: 5
Mul: 50
Div: 2

=== Code Execution Successful ===

4. **Create a class template to input and display array elements.**

```
#include <iostream>

using namespace std;

template <typename T>

class Array {

    T arr[100];

    int n;

public:

    void input(int size) {
```



```

        n = size;

        for(int i = 0; i < n; i++)

            cin >> arr[i];

    }

    void display() {

        for(int i = 0; i < n; i++)

            cout << arr[i] << " ";

    }

};

int main() {

    Array<int> obj;

    obj.input(3);

    obj.display();

}

```

```

input
1 2 3
1 2 3
...Program finished with exit code 0
Press ENTER to exit console.

```

